

Housing Market Dynamics and the Future of Housing Prices*

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I. INTRODUCTION

Owner-occupied housing today constitutes the largest single source of wealth in the United States. Clearly, the future of that wealth is not just of intrinsic interest but also has important implications for the nation's macroeconomy. In a recent paper, Mankiw and Weil [26] have suggested that after many decades of real value increases, residential home prices in the next 10 years should experience significant real value declines. This reversal is predicted because of the future demographic profile of the country in which slower household formation and an older age distribution act as negative shocks to housing demand. This has led to some popular speculation about an impending real estate collapse and its ensuing economic implications [4, 30].

Our view is that much of this concern is misplaced. With a well specified macro-model of the single family housing market, demographic forces should only slow the appreciation of American homes, not generate significant price decreases even in constant dollars.

To arrive at this conclusion, we develop a structural model of the single family market that evolves from a long tradition of macroeconomic housing research. Our model is a variation of the stock-flow approach and contains several innovative features. First, we question the traditional

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stock-flow assumption that the housing market clears quickly. Using an explicit test of how rapidly house prices adjust to equilibrium, we find strong evidence that it takes several years for the market to clear. This inability of prices to rapidly adjust both supports and helps to explain some of the recent evidence on the inefficiency of the housing market [8, 18]. Second, we estimate our equations using two quite different approaches to the issue of how consumers form expectations about future house prices. The statistical superiority of gradual price adjustment holds strongly both when consumers develop expectations by looking backward at historic price movements as well as when housing demand is based upon rational forward-looking forecasts.

Our final innovation involves the behavior of single family housing construction—clearly as relevant to any forecast of future prices as changes in demand. In most studies of housing construction, the production or flow of new units is determined by the level of house prices. In contrast to these studies, we view the supply of housing as more closely intertwined with the factor market for land. Urban spatial theory defines a long run rising supply schedule for the stock of housing units. The average price of new units increases as this stock grows, because usable land becomes more scarce. Under this theory, high price levels will generate an increased flow of units only until the current stock catches up with this long run supply schedule. Our econometric results strongly suggest that models with price levels work better when lagged stock or other measures of stock disequilibrium are included. These models also yield clear evidence of an elastic long run stock supply schedule.

Our forecasts suggest that the nation's changing demographics will impact the demand for housing in several different directions. On net, the changing demographics of the 1990s do not have as severe a negative impact on housing demand as previously thought. The more important point, however, is that with price elastic construction in the short run and a rising long run supply schedule for the stock of housing, it is impossible for prices to undergo any significant and sustained decline.

While many of these theoretical issues might best be explored with data about individual metropolitan housing markets, we have chosen to use aggregate data so as to be able to address the issue of future price movements in light of the nation's changing demographics. The next section of the paper reviews previous approaches to macroeconomic modeling of the housing sector, while Section III discusses the theoretical arguments in favor of our various innovations. Sections IV and V review the data on price movements, look at the nation's changing demographics, and present our model estimates of housing price determination. Section VI then examines the cyclic behavior of housing starts and presents alternative econometric results for a housing supply equation. Putting the

models together in Section VII, we generate several forecasts of the housing market and estimate what movements in prices should occur with and without the impending demographic changes.

II. THE LEGACY OF HOUSING MARKET RESEARCH

Aggregate models of the housing sector largely have their origins in the multi-sector macroeconomic models that evolved during the 1960s. The initial objective of such efforts was mainly to forecast the level of new residential investment [1, 6, 25]. Gradually, interest shifted more to the owner-occupied housing sector with an additional focus on price movements and the role of financial institutions or credit markets. More complete models were then developed with demand as well as supply equations [21, 41].¹ Almost without exception, these early research efforts estimated part or some variation of a basic two-equation stock-flow model of the housing sector.

In these early models, the demand for (owner-occupied) housing depends on exogenous variables X_1 (such as demographic characteristics and real permanent income), the real price level of housing, P , the annual user cost of financing that price, U , and the alternative cost of renting, R . Until the last decade, this financial user cost of homeownership considered only mortgage interest rates and not the tax deductibility of these payments or expected capital gains from house price appreciation.

The traditional assumption in the stock-flow model is that markets clear quickly and, at any time, prices adjust to equate the demand for housing with the existing stock (Eq. (1)). Thus, equilibrium price levels are determined in each period as a function of the housing stock, demand instruments, and mortgage interest rates.²

The supply of housing, on the other hand, follows the differential equation (2), where the stock slowly depreciates (at a rate δ) and expands gradually with new construction, C . Construction, in turn, is always assumed to depend on housing prices, P , and exogenous variables, X_2 . Generally included in X_2 are factor costs and various interest rates.

$$D(X_1, P, U, R) = S \quad (1)$$

$$\Delta S = C(X_2, P) - \delta S. \quad (2)$$

¹ Much of this early work is summarized in Fair [15].

² In theoretical models with no capital market constraints, the demand for owner-occupied housing should depend only upon the relative annual cost of owning as opposed to renting: PU/R [36]. In most empirical studies, however, demand has depended separately upon price levels, interest rates, and average rents (P, U, R). This more flexible formulation allows for the facts that because of borrowing or downpayment constraints housing prices may exert a separate impact on demand and that current homeowners with fixed rate mortgages are not exposed to upward interest rate risk.

If S is measured in units, then the demand function in Eq. (1) reflects household formation decisions as well as tenure choice. Alternately, if S is measured in dollars, then the demand function combines decisions about not only household formation and tenure choice but also the consumption of housing services. Early estimates of the supply equation (2) were made using data on both housing starts and the value of new investment. Most estimates of the demand equation (1), however, have been cast in terms of housing units. The number of housing units is known with some precision while estimates of the value of the nation's housing stock exclude land and include only capital.

In the early 1980s, an important contribution to the theory of housing demand was made with a more precise definition of the user cost of homeownership. Applying life-cycle models to the consumption of housing, a number of authors suggested that the correct measure of user cost should be an after-tax inflation-adjusted measure. First, the term (U) incorporated the after-tax cost of debt and property taxes $(i + t_p)(1 - t_y)$. Here, i is the nominal interest rate and t_p and t_y are marginal tax rates on property and income. The inflation adjustment added the expected rate of future nominal price appreciation for housing assets $E(\Delta P/P)$. There was considerable discussion about the financial market conditions which would be necessary for consumers to regard expected future capital gains as equivalent to current mortgage outlays [13, 39, 45]. Modern demand theory now recognizes that the user cost of housing must be based on an equation such as (3), which because of the expected price term becomes an endogenous variable in the model.³

$$U = (i + t_p)(1 - t_y) - E(\Delta P/P). \quad (3)$$

Kearl [24] is the first author to incorporate Eq. (3) into Eq. (1) and solve for house prices as a function of demand instruments, the housing stock, mortgage rates, and expected future price inflation. Poterba [31] formulates a similar model but statistically estimates only an equation for housing investment and not prices. Most recently, Mankiw and Weil [26] also use a model that incorporates Eq. (3) but estimate the demand equation without including the stock of housing.

A final extension of housing stock-flow models involves the tandem modeling of related factor or financial markets. The mortgage interest rate, for example, is sometimes made endogenous and determined simul-

³Life-cycle models require that for (3) to hold, borrowing constraints must be either absent or non-binding. In addition, the expected price term refers to current or next period price inflation only if there are no transaction costs to altering housing consumption. When transaction costs impede mobility, the price term must consider planned holding periods.

taneously by housing demand and interest rates in the rest of the economy [22, 41]. Likewise, Smith [41] suggests that housing construction costs might be endogenous and models them as depending upon the level of home building activity and construction costs elsewhere in the economy. More recently, Topel and Rosen [43] assume that housing suppliers recognize the rising supply schedule for construction and phase in new building so as to minimize intertemporal costs.

While this history of stock-flow models has spanned almost three decades, the empirical results that have emerged are remarkably consistent in yielding several important conclusions about the operation of the housing market.

The housing market has a somewhat predictable cycle with positive serial price correlation. A number of recent studies have demonstrated the inefficiency of the owner-occupied housing market with the observation that house prices are far from a "random walk" and move in smooth, predictable patterns [8, 18, 26]. Consistent with this behavior is the observation that the market-clearing price equations in earlier stock-flow models usually exhibit significant autocorrelation (e.g., Smith [41]). Such serial correlation, both in prices and in their error structure, is likely an indication of more complicated price dynamics than those that emerge from a simple moving market equilibrium [17].

The housing market may exhibit significant disequilibrium so that prices are not a sufficient statistic. Eubank and Sirmans [14], Rosen and Smith [37], and Fair [15] have all produced evidence that prices or rents move only gradually, and that vacancy rates operate like a Walrasian adjustment mechanism. On the supply side of the market, estimated models of new housing construction invariably contain a range of variables in addition to house prices and construction costs. Vacancy rates, mortgage rates, sales rates, and general macroeconomic conditions all seem to add significantly to equations forecasting building activity. If the market were clearing rapidly and in equilibrium, such factors would be fully incorporated into house prices which then would be the sole determinant of new supply.

The connection between housing construction and the various factor markets remains largely elusive. Most housing investment equations include the ratio of house prices to a construction cost index as an independent variable. Usually, such variables have been significant although only at relatively low confidence levels. Poterba [31] and, more recently, Topel and Rosen [43] are the first to use cost indices separate from house prices. Neither is able to obtain a significant cost coefficient. The relationship between housing construction and the land market has been virtually

unexplored in time series research largely because of very limited data on land prices.

III. SOME REVISIONS TO HOUSING MARKET STOCK-FLOW THEORY

Given the lessons of housing market research, the structure of the traditional stock-flow model might benefit from some revision. This would seem particularly appropriate in light of the considerable microeconomic theory about housing and land markets that has been developed over the last 10 to 15 years. We suggest that three innovations would help to make the model more consistent with current microeconomic knowledge.

Gradual price adjustment. The stock-flow model is predicated on the assumption of gradual supply adjustment. There is compelling evidence that the demand side of the market also does not clear quickly. The fact that housing prices are incomplete predictors of new construction, the observation that vacancy rates generate gradual price changes, and the observed tendency for positive serial correlation in housing prices all can be explained if prices adjust only gradually over a number of periods in response to shocks.

Explanations for slow market clearing are easily found in the recent literature on consumer search in housing markets. With product heterogeneity and time-consuming search, the anticipated sales time for a house can be long and exhibit significant variance [10, 19, 46]. Individual sellers cannot easily determine if an overly long sales time represents an unanticipated downturn in the market or simply random misfortune. Within this environment, rapid price adjustments may not be rational [33, 38, 42].

Gradual market clearing can easily be incorporated into the basic stock-flow model with the addition of a price adjustment mechanism. If the solution to Eq. (1) is a hypothetical equilibrium price P^* and τ is the (annual) percentage rate at which actual prices converge to this equilibrium price, we can add Eq. (4) to the model (1)–(3).

$$\Delta P = \tau[P^* - P]. \quad (4)$$

Expectations. The expanded model, now encompassing Eqs. (1) through (4), needs a definition of how price expectations are formed to be complete. Sheffrin [40] and then Poterba [31] adopted the macroeconomic theory of rational expectations, assuming that forward forecasts use all the information available about a market's operation. In effect, the expected value of future prices is the solution to the model when future exogenous variables are uncertain. While markets that operate with rational expectations are held to be efficient, prices can still exhibit systematic patterns

and serial correlation. If the exogenous variables to the market move systematically, or if the market itself permits only gradual adjustments to its stock (such as in Eq. (2)), then even rationally forecasted price movements can be highly correlated over time.⁴

An additional source of direct intertemporal price dependence can be introduced into the model if expectations are formed from distributed lags of previous price movements. Two recent studies of the housing market [8, 26] provide some strong evidence that such extrapolations are a common (albeit irrational) form of household behavior. With backward-looking expectations, then, current prices will be positively related to recent past prices, providing a further reason for serial price correlation.

Housing prices, new construction, and the land market. In most models of housing construction, the homebuilding industry is composed of competitive firms who face rising factor cost schedules. In the short run, rising costs result from bottlenecks and, in the long run, from increasing supply schedules for labor and raw materials. Thus, a long run increase in house price levels leads to a *permanent* increase in the flow of construction, the magnitude of which depends upon how much factor costs increase with the greater level of building activity.

The problem with this view is that one of the most important factors in the supply of housing is land and land prices depend on the *stock* of housing, not the flow or level of building activity. With this kind of factor supply schedule, the impact of an increase in housing prices is quite different than that in the traditional model described above. The rise in prices initially generates excess returns, but the flow of construction increases only *temporarily* (above some “normal” level). As the stock of units grows, land prices rise and eventually absorb the excess returns. Construction then declines back to its normal level. In this model, the price level determines the long run stock of units. Construction responds to an increase in prices only temporarily, until the existing stock catches up to the longer run equilibrium stock.

Urban spatial theory has dealt in great detail with the relationship between the land market and the stock of housing. Ricardo [34], Alonso [2], Mills [27], Muth [28], and Wheaton [44] have all developed static theories of the urban land market, in which intra-urban transportation costs generate rent and density gradients. In equilibrium, land is developed (for housing) until the urban land rent gradient intersects agriculture

⁴In Poterba [31], a positive shock to demand generates a fully anticipated sudden price increase and then gradual (serially correlated) decreases in price as the anticipated supply is slowly added to the stock. In Mankiw and Weil [26], a baby boom shock to the market, once revealed, is shown to generate a long gradual increase in prices since the revealed births occur many years before the fully anticipated impact on the market.

rent at the city's edge. In a comparative static analysis, as urban areas grow, land rents must rise throughout the city in order to expand the edge of development and bring forth the land needed for new housing. Recent spatial theory has tried to more explicitly characterize such adjustments to the housing stock. Anas [3], Fujita [16], Wheaton [45], and Braid [7] have all developed models of urban growth in which housing capital is durable. This work only reinforces the conclusion that land prices increase significantly with the stock of housing units.

To date, spatial theory has used only equilibrium models, in which the stock of housing always equals the urban population. Thus, by assumption, the flow of housing (new construction) equals the growth in population. In effect, there is no independent supply theory dealing with construction flows or the conversion of land for urban development. A simple and quite straightforward model of housing construction can be created by combining a stock adjustment process with a long run spatially based definition of the equilibrium housing stock. We denote $S^*(X_2, P)$ as the long run stock of housing that emerges from the spatial theory described above, where X_2 represents an array of cost shifters (as in Eq. (2)). If α represents the speed with which the stock adjusts through new construction, we can replace Eq. (2) with (2') below.⁵

$$\Delta S = C - \delta S = \alpha [S^*(X_2, P) - S] - \delta S. \quad (2')$$

The difference between (2) and (2') is the inclusion of current or lagged stock levels in an equation predicting construction. In (2'), housing price levels generate new construction only if those prices dictate a value of S^* that differs from the current stock. Thus, a large city experiencing no growth may have high land and housing prices (from Ricardian spatial theory), but little or no construction since the city's stock has adjusted to equilibrium levels. Conversely, a much smaller but rapidly growing city

⁵One implication of a stock adjustment process is that the adjustment rate (α) reflects the average ratio of investment-to-stock. Since annual housing construction is only a small fraction of the stock, we expect a very small (α) value, in the 0.02 range. Differentiating Eq. (2') yields

$$\frac{\partial C}{\partial P} \frac{P}{C} = \left(\frac{\alpha S}{C} \right) \frac{\partial S^*}{\partial P} \frac{P}{S}.$$

In a long run steady state ($S = S^*$), the elasticity of the stock with respect to price must equal that of construction. Thus, in the expression above, $(\alpha S/C)$ must equal 1.0 and therefore α is the ratio of C/S . Muth [29, pp. 35–36] put forth a demand model for the housing stock in which the stock also adjusts gradually towards a long run desired equilibrium value.

could have lower price levels together with significant construction as S^* is rapidly shifting and leading the current stock.

We believe that the theory of one city's land market can be appropriate for explaining movements in aggregate national housing construction. Over the last 30 years much of the growth in the nation's single family housing stock has occurred from construction at the edges of its more rapidly growing cities. In the long run, then, it is the increase in and not level of house prices that brings forth the land for such growth.

IV. DATA SERIES NATIONAL DEMOGRAPHIC TRENDS

Ideally, an aggregate housing model should consider the demand and supply for total housing services. With the exception of the recent paper by Mankiw and Weil [26], however, past stock-flow models have been cast almost exclusively in terms of housing units. In large measure this reflects the relative ease with which the number of units can be tracked and the difficulty of determining the aggregate value of the nation's housing stock. The Commerce Department does produce an estimated series on the value of the residential capital stock but it excludes the value of land.⁶ Developing a model of housing services using this series would ignore the important role played by the land market. Alternatively, by modeling the market for housing units, we are able to use several new transactions-based data on quality-controlled housing prices. These, of course, include a significant land component.

In examining alternative sources of national data on housing prices, the most troublesome issue is finding a true quality-adjusted house price index. Average resale prices for existing homes have often been suggested as a proxy for a price index since the stock changes only slowly over time. Using data from four individual cities, however, Case and Shiller [9] find that these stock changes may not be so gradual. In addition, they argue that seasonal and cyclic factors can also influence the composition of sales, which casts further doubt on the ability of average resale price data to reflect a true price index.

Currently, two national price series are available that do control for the quality of units sold: the Commerce Department's price index for newly constructed single family homes and the recently released Freddie Mac series for existing houses. The Commerce Department's price series uses an hedonic approach and goes back to 1963. The Freddie Mac series is based on a large sample of repeat sales and uses a match sales methodol-

⁶Mankiw and Weil [26] use the stock of residential capital, constant cost valuation, from *Fixed Reproducible Tangible Wealth*. Capital stock is defined as reproducible tangible wealth which does not include non-reproducible tangible wealth such as land and other natural resources.

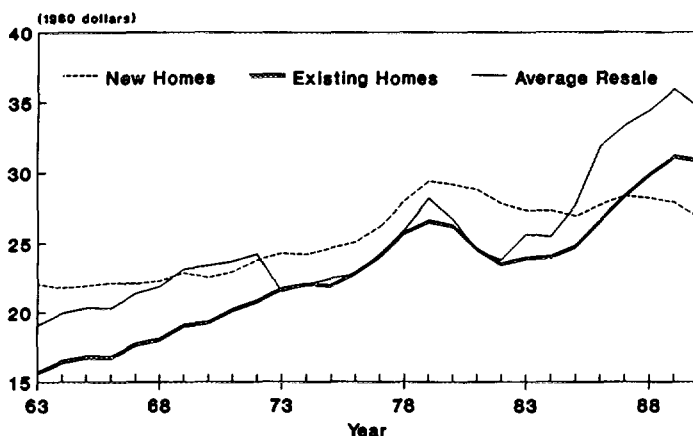


FIG. 1. Alternative house price indices. New homes and existing homes are constant quality indices.

ogy, but goes back only to 1970. A third series, from the Federal Home Loan Bank Board (FHLBB), covers existing units back to 1963 but does not control for quality changes. All three data series are plotted in Fig. 1, in constant (1960) dollars. The two series for existing house prices display quite similar long run growth as well as a consistent and smooth cyclic pattern. The decline in prices during recessions and their growth in between seem to reinforce the conclusion of price predictability and positive serial correlation that has been noted by other authors. The new house prices series, on the other hand, shows both slower long term growth as well as less cyclic movement. Since new homes represent only a small portion of the total housing market, an existing house price series is, in principle, preferred. Given that the two existing house series move quite closely, we have joined them, using the Freddie Mac quality-adjusted series from 1970 to 1990 and the FHLBB data for 1963–1969. For comparison we estimate our models both with this hybrid series as well as with the new house price series.

The demand for owner-occupied housing also depends on the annual cost of its only substitute, rental housing. The rent data used in this paper is the Bureau of Labor Statistics rent index which is a component of the consumer price index.⁷

⁷The Commerce Department repeatedly surveys several hundred thousand units nationally. This procedure produces an index which is felt to slightly understate true rental growth due to unit depreciation. We correct for this problem by applying a constant depreciation rate (see DiPasquale and Wheaton [12]).

The stock series used in Eqs. (1)–(4) should represent the potential number of units that are available for owner occupancy. Since only 6% of the owner-occupied stock is in multifamily structures and considering that our price series is for only single family units, we chose to ignore condominiums in our stock series. We began that series with the 1960 Census count of total single family units (detached and attached) minus single family units that are renter-occupied. To that, we added annual data on single family starts from the previous year. For 1970, we compared the calculated series with the new Census count and then repeated the process for 1970–1980 and 1980–90. For the final period, the benchmark used was from the 1987 American Housing Survey (AHS). This procedure, of course, will overstate increases in the stock since it ignores changes due to demolition and abandonment. In order to correct this problem, annual removal rates were applied so that the calculated stock series would meet the census benchmark each decade. Reasonable annual removal rates of 0.35, 0.30, and 0.20% were required to reconcile the annual stock series with the census and AHS benchmarks.⁸

The main endogenous determinant of housing demand beside house price levels is the user cost of homeownership. Our measure of this takes federally deductible mortgage interest and property tax payments, to which we added economic depreciation and then subtracted estimates of expected future price appreciation in housing. The user cost variable was calculated with both rational and backward-looking price expectations. The former uses an instrumental variable forecast of current period price inflation where the instruments are historic values of the model's exogenous and endogenous variables. The latter computes a three-year moving average of past house price inflation.⁹

To capture the exogenous determinants of housing demand, we used personal consumption, a standard measure for permanent income, and

⁸There is some concern that the Census Bureau changed its definition of "units in structure" between 1960 and 1970. Single family units are defined to include both detached and attached units. The 1960 count of 3.7 million attached units drops to 1.9 million in 1970. From conversations with Census officials we conclude that the 1960 definition of "attached" led to an overcount of these units. In addition, the 1960 count includes seasonal units which are not included in the 1970 or 1980 counts. As a result, we adjusted the 1960 count of units to accommodate the estimated overcount as well as the seasonal units. These adjustments bring the growth in the stock from 1960 to 1970 more in line with the number of starts and the growth in households over the decade.

⁹The mortgage rate used is the Federal Home Loan Bank Board contract rate. Marginal tax rates were calculated for the income decile that characterized first time homebuyers in the 1985 AHS [11]. That decile was then used in all other years, together with each year's tax schedule and estimated housing and non-housing deductions, to arrive at the appropriate tax rate. In general, the marginal rate ranged from 19% in 1960, to 24% in the early 1980s, to 15% in 1990.

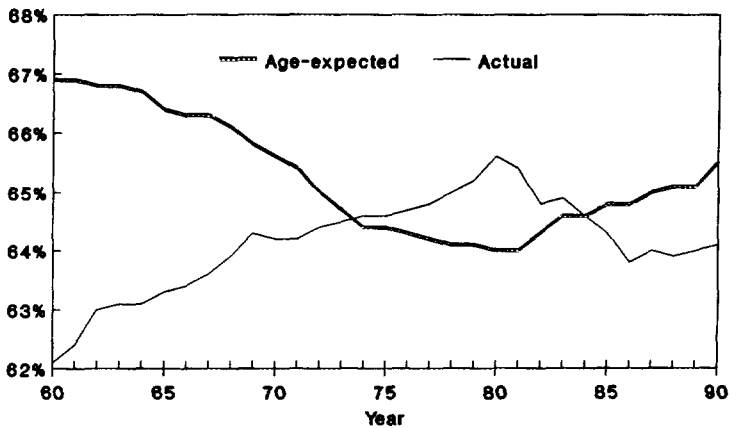


FIG. 2. Actual vs age-expected homeownership rates.

calculated two demographic variables. The first is the fraction of households who normally would be expected to own homes.¹⁰ This calculated variable takes a given set of homeownership rates by age (in this case those in 1973) and then uses the annual age distribution of households to estimate the age-expected homeownership rate.¹¹ In effect, this variable isolates the likely impact on homeownership of changes in the nation's age distribution. As shown in Fig. 2, our age-expected rates move almost inversely to actual homeownership rates. The actual rate increased steadily from 1960 to a peak of 65.6% in 1980, and then dropped to 64% where it stands today. By contrast, our age-expected ownership started to fall in the 1960s as baby boomers began to form households. It bottomed out in the early 1980s and with a maturing population will continue to rise well into the next century. Thus, the age distribution in the country acted as a *negative* shock to the demand for single family housing during the 1970s and has provided some *positive* stimulus in recent years.

Our second demographic variable is household formation. As shown in Fig. 3, the net increase in households rose noticeably during the 1970s and fell gradually in recent years. Several authors [5, 20] have argued that

¹⁰Our calculation of age-expected homeownership is similar to that developed by Jaffee and Rosen [23]. Our calculations use a more detailed age distribution but do not consider different family types.

¹¹Varying the age-specific ownership rates by using data for a different year shifted the calculated age-expected variable slightly, but left its pattern over time largely unchanged.

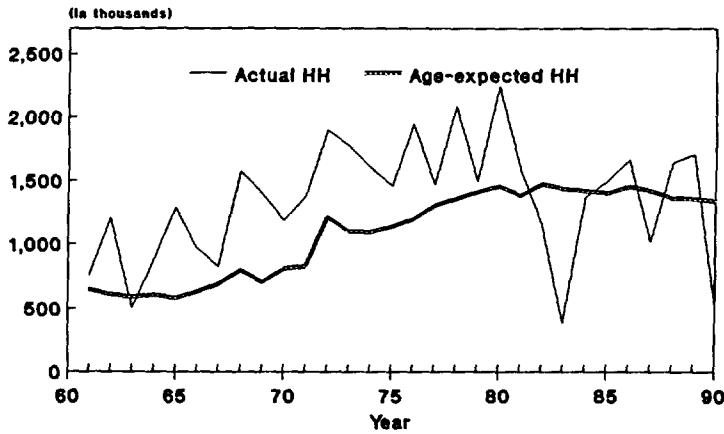


FIG. 3. Actual vs age-expected household formation.

household formation decisions are influenced by real housing prices, a fact which could make the number of households an endogenous variable in any model of the housing market. To allow for this possibility we also created an exogenous age-expected number of households. The calculation of the age-expected household variable is analogous to the calculation of the age-expected homeownership discussed above. We took the headship rates by age in 1973 and weighted them by the age distribution of the population for each year. As illustrated in Fig. 3, a comparison of the two formation rates shows that during much of the 1960s they moved similarly. From 1968 to 1980, however, actual household formation ran way ahead of that predicted by the age distribution. By contrast, in the 1980s, actual household formation lagged slightly behind the age-predicted variable.

The exogenous variables chosen to predict new construction were based largely on the accumulated experience of past housing studies. We relied on the Commerce Department's construction cost indices for labor and materials and then weighted them into a single index. We used the price series for farm land provided by the Department of Agriculture. Previous studies have found significant impacts by short term real interest rates, so we used a three-month Treasury bill rate minus a moving average of CPI inflation over the last three years.

Past studies of residential building have found that, in addition to house prices and construction costs, other indicators of current or potential market conditions add appreciably to a supply equation. Such variables have included vacancy rates [25, 35], months on the market [43], or movements in mortgage rates [41]. As discussed previously, such variables

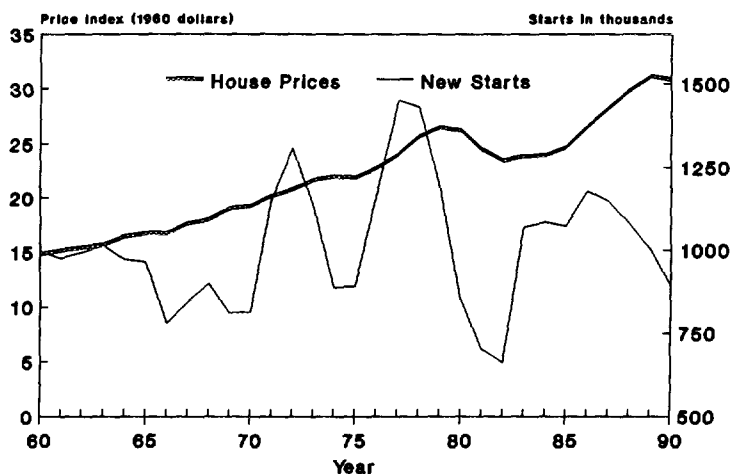


FIG. 4. House prices and new single family starts.

matter presumably because house prices are slow to adjust, and thus are incomplete measures of the market. This can clearly be seen in Fig. 4, where house prices are graphed against new single family starts. While real house price levels were roughly the same in 1982 as in 1977, starts reached a record high of 1.45 million in 1977 and a record low of 663 thousand in 1982. Although short term real interest rates were higher in 1982, differences in construction of this magnitude were more likely the result of the very different economic outlooks at the time. For leading indicators of the housing market, we use the median months on the market for new units recently sold and the current rate of national employment growth.

It is important to note that many of these data series are not available at the subnational level. The annual series on the age distribution, number of households, and constant quality house prices, for example, have not been disaggregated by metropolitan areas within the United States. This provides an obvious additional reason for estimating models with national data.

V. DEMAND ESTIMATION AND PRICE DETERMINATION

Our specification of the underlying housing demand equation (1) uses a functional form that provides for an interaction between the number of households and other demand variables. The term in parentheses in Eq. (5), therefore, can be thought of as representing the fraction of households

that desire to own single family units:¹²

$$H_t(\beta_1 R_t + \beta_2 \text{OWN}_t + \beta_3 \text{WAGE}_t + \beta_4 P_t + \beta_5 U_t) = S_t, \quad (5)$$

where H is total households, R is the rent index, OWN is the age-expected homeownership rate, WAGE is the permanent income per household, P is the price index of single family housing, U is the annual user cost of homeownership, and S is the stock of single family units.

Rearranging Eq. (5), we solve for the market clearing price, P_t^* , as a function of demand determinants and the ratio of stock to households,

$$P_t^* = (1/\beta_4)[S_t/H_t - \beta_1 R_t - \beta_2 \text{OWN}_t - \beta_3 \text{WAGE}_t - \beta_5 U_t]. \quad (6)$$

As suggested in Section III, even with annual data there is substantial evidence that the house prices move slowly and that the market may not clear within one period. Gradual market clearing is incorporated into our stock-flow model with the addition of a price adjustment mechanism. Denoting τ as the (annual) percentage rate at which actual prices converge to the equilibrium price P_t^* , we have the lagged price model (7). Since market clearing (Eq. (6)) is simply a special case of gradual adjustment (Eq. (7)), the significance of the parameter τ is a clear, nested hypothesis test of how quickly the market clears.

$$P_t = \tau P_t^* + (1 - \tau)P_{t-1}. \quad (7)$$

To estimate the combined Eqs. (6) and (7), we have used a number of data and methodological variations to test the robustness of our analysis. Our first set of results uses the series on actual households and is presented in Table 1. All equations are estimated by two-stage least squares, because of the likelihood that current rents are endogenous.¹³

¹²The interaction between households and the other variables not only makes sense theoretically, but produces better econometric results than a straight linear specification. The term within parentheses represents tenure choice when H_t is measured as actual households and a combination of both tenure choice and household formation when H_t is measured as potential age-expected households.

The variables within parentheses could in principle be entered multiplicatively, and then the entire expression estimated log-linearly. However, our measure of the user cost variable, like that of other researchers, is negative during the late 1970s. Since non-linear models are not scale invariant, scaling the variable to avoid negative values alters the estimates of all parameters, leaving the results dependent on the chosen scalar.

¹³Instruments are lagged endogenous variables and current period exogenous variables. To ensure identification, the lagged stock of multifamily rental units was also included.

TABLE 1
Demand Equations for Actual Households

	Dependent variable:		Existing house price		New house price
	Equation 1	Equation 2	Equation 3	Equation 4	Equation 5
S_t/H_t	-114.40 (-1.12)	-89.89 (-2.71)	-122.53 (-1.63)	-65.60 (-5.10)	-60.22 (-1.67)
$WAGE_t$	6418 (6.14)	2346 (5.20)	6488 (7.10)	1518 (6.30)	1009 (2.08)
OWN_t	82.88 (0.86)	66.28 (2.14)	92.24 (1.30)	47.93 (3.95)	53.02 (1.56)
U_t	-0.14 (-1.43)	-0.15 (-4.78)	—	—	—
U_t (IV)	—	—	0.06 (0.85)	-0.13 (-8.86)	—
P_{t-1}	—	0.71 (13.98)	—	0.84 (26.87)	—
R_t (IV)	-0.39 (-1.40)	-0.03 (-0.30)	-0.44 (-1.65)	-0.001 (-0.012)	-0.04 (-0.27)
U_t (New house price)	—	—	—	—	-0.20 (-3.91)
P_{t-1} (New house price)	—	—	—	—	0.83 (12.37)
R^2	0.931	0.993	0.928	0.998	0.976
D-W	0.520	1.793	0.647	1.414	1.179

Note. All equations are estimated using two-stage least squares. U_t uses backward-looking expected existing house price appreciation; U_t (IV) uses "rational" forecasts of future existing house price appreciation; U_t (new house price) uses backward-looking expected new house price appreciation. R_t (IV) is an instrumental variables estimate of rent. t statistics are in parentheses.

Very similar equations are obtained using lagged rents with OLS estimation. The first two columns give the market clearing equation (6) and then the adjustment equation (6)–(7) when the annual user cost of ownership is based upon backward-looking price expectations. The third and fourth columns repeat these equations when the annual user cost is based upon instrumental variables estimates of house price inflation. The final column gives an adjustment equation using the new house price data series. Table 2 is identical to Table 1 but uses the calculated age-expected households in place of actual households.

The results in Table 1 tell a clear and consistent story. The addition of price adjustment to the market clearing model (column 2 versus column 1), increases the R^2 , and significantly changes coefficients. The estimated value of τ lets prices move to 29% of their long run equilibrium value in the first year following a shock.

TABLE 2
Demand Equations for Age-Expected Households

	Dependent variable:		Existing house price		New house price
	Equation 1	Equation 2	Equation 3	Equation 4	Equation 5
S_t/H_t	14.29 (0.20)	-57.23 (-2.35)	-64.25 (-0.97)	-54.60 (-3.85)	-43.80 (-1.80)
$WAGE_t$	3287 (1.51)	2788 (3.72)	5654 (2.71)	2269 (4.79)	1507 (2.11)
OWN_t	-39.52 (-0.57)	37.64 (1.56)	39.45 (0.61)	39.38 (2.83)	37.27 (1.64)
U_t	-0.14 (-1.65)	-0.14 (-4.57)	—	—	—
U_t (IV)	—	—	0.06 (0.90)	-0.11 (-6.67)	—
P_{t-1}	—	0.71 (13.03)	—	0.84 (21.18)	—
R_t (IV)	0.32 (0.59)	-0.15 (-0.81)	-0.29 (-0.56)	-0.19 (-1.71)	-0.17 (-0.92)
U_t (new house price)	—	—	—	—	-0.20 (-3.91)
P_{t-1} (new house price)	—	—	—	—	0.88 (11.25)
R^2	0.942	0.993	0.936	0.997	0.976
D-W	0.387	1.859	0.590	1.699	1.200

Note. All equations are estimated using two-stage least squares. U_t uses backward-looking expected existing house price appreciation; U_t (IV) uses "rational" forecasts of future existing house price appreciation; U_t (new house price) uses backward-looking expected new house price appreciation. R_t (IV) is an instrumental variables estimate of rent. t statistics are in parentheses.

The improvement in the model with the inclusion of a price adjustment term is even more pronounced when rational forecasts are used to compute the expected user cost variable (column 4 versus 3).¹⁴ Without gradual price adjustment, the coefficient on user cost has the wrong sign and many coefficients are insignificant. The rate of price adjustment is only 16% in the rational forecast model, which is considerably slower than that in the backward-looking version. Slower price adjustment probably results with rational expectations because it helps explain some of the market's serial price correlation, which in Eqs. (1) and (2) is accounted for by the assumption of backward-looking expectations.

¹⁴The rational expectations version of user cost uses an instrumental variables forecast of current house price inflation. The instruments are all current period exogenous variables and lagged endogenous variables.

The last column in Table 1 shows that a reasonable equation can be estimated when the constant quality new house price series is used as the dependent variable. It is perhaps not surprising that this model is slightly inferior to those using resale house prices since new houses are a more specialized submarket and make up roughly only 17% of each year's total transactions.

The story is very similar in Table 2 where the age-expected number of households, instead of actual households, is used as a demand determinant. The market clearing models continue to suffer from severe autocorrelation and the stock-to-household ratio and user cost variables are insignificant.¹⁵ On the other hand, the price adjustment models in Table 2 are similar to those in Table 1. The most important difference is that the price adjustment models with actual households (Table 1) have larger coefficients on stock and the age-expected ownership rate than do the adjustment models with age-expected households (Table 2).

Across the various equations with price adjustment (Eqs. (2) and (4) in both tables), the price elasticity of demand consistently is in the -0.09 to -0.19 range. This is more inelastic than that obtained by other studies, but those involve the impact of price on the demand for housing services. Given the small (5.6%) movement in the nation's homeownership rate over the past three decades, in contrast to a 110% swing in real house prices, it is reasonable to expect a much lower price elasticity for unit tenure than for services.¹⁶

It is important to compare the price elasticity of demand to the demand elasticities with respect to user cost and rent.¹⁷ In an unconstrained capital market, with full interest rate exposure, demand should depend only on the term PU/R . In this case, the elasticities of demand with respect to P , U , and R obviously are all identical. When evaluated at sample averages, the coefficients above yield far different impacts. The elasticity of U is only around -0.004 , or less than one-thirtieth of that for P . While the rent elasticity is sometimes close to that for price, it is never

¹⁵Following the tradition of Granger and Newbold [17], we interpret autocorrelation as an indication that the model's underlying specification is flawed, rather than an econometric problem to be fixed technically.

¹⁶As summarized in Quigley [32], cross-section studies place the demand elasticity for housing services with respect to housing price in the -0.5 to -0.7 range. These studies, however, estimate elasticities with respect to housing price levels only, and do not include a variable measuring the asset cost of homeownership. Among time series studies of housing unit demand, such as Rosen and Rosen [36] or Smith [41], only Kearn [24] undertakes any elasticity calculations. Kearn's price elasticity of unit demand is -1.5 .

¹⁷From Eqs. (5)–(7), the price elasticity is $\beta_4 PH/S$. The user cost elasticity is $\beta_5 UH/S$. The structural parameter β_4 is recovered as the ratio of one minus the estimated coefficient for lagged price over the coefficient for S_t/H_t . The structural parameter β_5 is recovered as the ratio of the coefficient for U divided by that for S_t/H_t .

significant statistically. Clearly, price levels are much more important than rents, interest rates, and price expectations in determining the demand for single family homeownership.

As for the exogenous determinants of unit demand, the elasticity with respect to permanent income is generally in the 0.3 to 0.7 range. These estimates again are somewhat below existing estimates of the income elasticity of housing services per unit.¹⁸ A percentage point change in the age-expected ownership rate should exert a similar change in the demand for housing units. With the adjustment models, the OWN coefficients in Tables 1 and 2 yield between a 0.7 and a 0.8 percentage point change.

It is interesting to note almost all elasticities are higher when age-expected households are used (Table 2) than when actual households (Table 1) are used. This is to be expected since when age-expected households are used the term within parentheses in Eq. (5) represents not just tenure but also household formation decisions. The impact of house prices on the combined decisions should clearly exceed that on tenure alone.

VI. CONSTRUCTION AND THE LONG RUN SUPPLY OF HOUSING

Our initial supply equation estimates single family housing starts (C_t) as a function of current house prices (P_t), an array of cost shifters, and the stock of housing (S_{t-1}). The cost shifters included in our model are the real cost of short term construction financing ($TREAL_t$) and cost indices for construction ($COST_t$) and land ($FARM_t$). Using a simple linear form, the estimating equation is (8), where α is interpreted as the annual rate with which the stock adjusts to its long run desired level,¹⁹

$$C_t = \alpha_1 + \alpha_2 P_t + \alpha_3 TREAL_t + \alpha_4 FARM_t + \alpha_5 COST_t - \alpha S_{t-1}. \quad (8)$$

Our estimate of Eq. (8) is presented in the first column of Table 3. The price and stock coefficients have their expected signs, but neither is statistically significant. The cost shifters all have the expected negative signs but only the real short term interest rate is statistically significant. The problem of insignificant cost indices is one that other studies have

¹⁸The income elasticity of aggregate housing services is the sum of the elasticities for units and for services-per-unit. Quigley [32] reveals some considerable consensus that the permanent income elasticity of housing services-per-unit demand lies in the range of 0.7 to 0.9. Kearl's study is the only time series model of housing unit demand that presents an income elasticity; in his model it is a very inelastic 0.25 [24].

¹⁹Experiments with log-log or log-linear functions yielded somewhat inferior results to this linear model.

TABLE 3
Supply Equations

	Dependent variable: Housing starts			
	Equation 1	Equation 2	Equation 3	Equations 4
Constant	1086.82 (0.93)	1138.39 (9.73)	649.80 (1.00)	-906.55 (-2.40)
P_t	67.29 (1.72)	57.48 (6.07)	47.40 (2.98)	—
$TREAL_t$	-72.60 (-4.88)	-59.17 (-9.44)	-53.53 (-6.56)	-54.44 (-6.28)
$FARM_t$	-0.28 (-0.16)	—	-0.39 (-0.42)	-3.28 (-5.00)
$COST_t$	-3.34 (-0.06)	—	27.78 (0.82)	66.88 (2.61)
S_{t-1}	-0.03 (-1.13)	-0.02 (-4.53)	-0.02 (-1.56)	-0.005 (-0.63)
$DEMP_t$	—	0.05 (3.40)	0.04 (1.96)	0.02 (1.18)
$SALE_{t-1}$	—	-84.24 (-3.25)	-113.64 (-3.19)	-119.11 (-3.29)
P_t (new house)	—	—	—	69.60 (2.74)
R^2	0.671	0.925	0.931	0.927
D-W	1.243	1.851	1.984	2.169

Note. All equations are estimated using OLS. t statistics are in parentheses.

also noted. For a time series model, the overall fit as measured by the R^2 is quite disappointing.

In the third equation in Table 3, we added in two leading indicators of market activity: the current change in employment ($DEMP_t$) and the number of months on the market for new homes recently sold ($SALE_{t-1}$). These additional variables greatly improve the fit of the model. Both $DEMP$ and $SALE$ are significant with the expected signs, and the R^2 jumps dramatically. $FARM$ and $COST$ remain statistically insignificant; in the second equation of Table 3, we dropped these two variables with little change in results.

In the fourth equation, we estimated the model using the price series for new homes. New house prices might be expected to have more of an impact on new construction than existing house prices, and in fact, the new home price coefficient is somewhat larger. The cost shifters, $FARM$ and $COST$, also become statistically significant, but the coefficient for $COST$ continues to have the wrong sign. Finally, the lagged stock coefficient is much smaller and insignificant. Of course, the price of new homes, located mainly at the edge of cities, should not in principle be the

determinant of the long run stock. It is existing house prices, reflecting locational value, that drive the size of the housing stock.

The stock coefficient in this model represents the speed with which the stock adjusts through new construction. The magnitude of the coefficient implies a very slow stock adjustment process (2% annual in Eqs. (2) and (3)). As discussed earlier (footnote 5), this is to be expected given the fact that new construction is such a small portion of the total stock at any point in time; over our sample period, annual construction has ranged from 1.3 to 3.3% of the stock. Using the stock coefficients from the second and third equations together with those for price, the long run price elasticity of the stock is in the 1.2 to 1.4 range.²⁰ As expected, the construction elasticity for these same equations is quite similar, ranging from 1.0 to 1.2.

A troublesome feature with all of these supply models is the strength of the coefficient for SALE. Over our sample period, the average number of months that new houses spent on the market ranged from 2.7 to 5.5. The coefficient in Eq. (2) implies that this variation would alter construction by 236,000 units. This is almost half the year-to-year market variation in new building activity. In fact, much of the short run movement in housing starts is explained by cyclic movements in SALE together with DEMP and interest rates (TREAL). These results are quite consistent with those provided by Topel and Rosen [43] who argue that such coefficients are too large to reflect simply interest costs and sales delays. The importance of such variables clearly indicates that with slow adjustments, prices are not a sufficient statistic in the housing market.

Finally, our supply models have relatively large intercepts, implying that there is a significant floor for housing starts. Over the 30 years covered in this analysis, starts generally trended upward with average levels of 926,000 in the 1960s and 1.14 million in the 1980s. However, they never dropped to lower than 663,000. If we take 1990 values for all variables except house prices and use the coefficients in Eq. (2), the house price index would have to drop from 30.8 to 16.3, a 47% decline, in order to bring construction down to a stock replacement level of 128,000 units. Clearly, while other variables drive short run construction fluctuations, in the longer run, supply is dictated by house prices relative to the size of the stock.

VII. DEMOGRAPHIC CHANGE AND THE FUTURE OF HOUSE PRICES

As we discussed in the introduction, Mankiw and Weil [26] suggest that after decades of real price increases significant declines in real house

²⁰ In this linear form, the elasticity of construction with respect to price is $\alpha_2 P_t / C_t$. The price elasticity of the desired stock is $\alpha_2 P_t / \alpha S_t$.

prices are likely over the next decade. The demand and supply models estimated in this paper provide an opportunity to forecast both future single family construction and house prices. In this section, we present our forecasts and compare them to Mankiw and Weil [26].

For our forecasts, we used demand equation (2) in Table 2 and supply equation (2) in Table 3.²¹ All forecasts begin with a set of expected demographic changes likely to occur over the next decade. The Commerce Department has made several such estimates through the year 2010 for both the number of households and the age distribution of the population. Since we used a demand equation where households are endogenous, our forecast must be based on an estimate of future age-expected households. Taking the Census Bureau's series 18 forecast of the population by age, we generate a continuation of our age-expected household variable based on 1973 headship rates.²² Under this scenario, estimated household formations fall to around 1.1 million annually by 2000, which is considerably lower than the 1.6 million annual level that characterized much of the 1970s. The age-expected homeownership rate, on the other hand, continues to rise, reaching 68% in 2000 after bottoming out at 64.0% in 1980-1981.

In the base forecast, we assumed smooth but slow growth in the economy using the long term employment and income forecasts from the WEFA Group, Inc. Income is projected to grow by around 1.8%, and employment by 1.3%, annually. We also assumed that the current monetary environment holds over the forecast period with 6% short term rates, 9% mortgage rates, and annual increases of 4.5% in the CPI. Finally, any exogenous cost inflation on the supply side was assumed to be smooth, at the CPI inflation rate. The expected sales time for new homes initially was fixed at four months which is close to the sample average.

Table 4 provides this base forecast for both construction and house prices. In this forecast, housing construction jumps 36% from 1990 to 1991, continues to rise slightly in 1992, and then falls somewhat through the rest of the decade. Real house prices rise through 1996 and remain quite stable over the rest of the decade. By 2000, real prices are 6.7% above 1990 levels.

To evaluate the impact of the nation's changing demographic profile, we undertook two forecasts with alternative assumptions about the nation's rates of aging and of household formation. The first set of assumptions is embodied in the forecast labelled DEM1. Here, we incorporated the expected slowdown in the nation's rate of household formation, but did

²¹These two equations form a recursive system in which each period's predicted values of construction (price) are used as inputs in determining the next period's price (construction).

²²The Census Bureau views the Series 18 projection as the most likely.

TABLE 4
Forecasts of Construction and House Prices 1991–2000

	Construction (thousands of units)				House Prices (1960 dollars)			
	BASE	DEM1	DEM2	Cycle	BASE	DEM1	DEM2	Cycle
1987	1146	1146	1146	1146	28.30	28.30	28.30	28.30
1988	1081	1081	1081	1081	29.90	29.90	29.90	29.90
1989	1003	1003	1003	1003	31.10	31.10	31.10	31.10
1990	894	894	894	894	30.80	30.80	30.80	30.80
1991	1214	1210	1238	879	31.74	31.66	31.66	31.27
1992	1236	1225	1260	1137	32.29	32.08	32.19	31.56
1993	1200	1178	1222	1170	32.53	32.15	32.41	31.68
1994	1175	1142	1197	1187	32.63	32.04	32.49	31.79
1995	1146	1100	1166	1149	32.80	31.98	32.63	32.01
1996	1121	1063	1148	1115	32.82	31.77	32.75	32.16
1997	1103	1035	1141	1099	32.79	31.54	32.91	32.30
1998	1078	1003	1130	1067	32.76	31.36	33.12	32.41
1999	1055	977	1124	1051	32.75	31.27	33.41	32.52
2000	1039	958	1121	1039	32.85	31.28	33.75	32.70

not allow the population to age. In effect, by comparing this forecast to that in BASE, we isolate the impact of the aging of the population. Since homeownership rises with age, removing the aging of the baby boom constitutes a negative shock to single family housing demand. In Table 4, real house prices are consistently lower in the DEM1 forecast (relative to that in BASE), and by the year 2000 are 4.8% less. Similarly, construction is slightly lower over the decade of the 1990s, and this yields a stock that is 0.6% less by the year 2000.

In the forecast labeled DEM2, we continued to remove the likely aging of the population but also increased the future rate of household formation back to levels that are in the same range as those that occurred during the 1970s and early 1980s. In this forecast, we effectively remove all traces of the baby bust that followed the post-war baby boom. In comparison to DEM1, the DEM2 forecast evaluates the impact only of a continued high growth of households. In DEM2, by the year 2000, prices are 7.9% greater, construction 17.0% higher, and the stock 1.0% larger than those in DEM1. The comparison between the DEM2 and the BASE forecasts measures the net effect of aging and slower household formation. The slower household formation from the baby bust more than offsets the aging effect, leading to a net negative demand shock over the next decade. By the year 2000, prices in DEM2 are 2.7% greater than those in BASE, while construction is 7.9% higher and the stock 0.4% greater.

Clearly, as estimated by our models, demographic factors do play an important role in shaping the market for single family houses. The combination of slower household formation and an aging population will, on net, act as a negative shock to housing demand in the 1990s. The magnitude of the shock, however, is relatively small. Furthermore, the long run supply of housing is quite price elastic and this plays an important role in mitigating the effect of any negative shocks to the demand side of the market. Mankiw and Weil [26] suggest that as a result of changing demographics, real house prices could drop by as much as 47% by 2007, and 30% over the next decade. To obtain such results with the models estimated here would require an enormous shock. For prices to drop 30% by 2000 relative to the base scenario, long run supply (and hence equilibrium demand) would have to shrink by 36%, given our elasticities. Since household growth is expected to be only 1.2% annually, even zero household growth for the next decade would not be a sufficient negative shock. In fact, dynamic experiments with the model reveal that a combination of no future household or income growth would reduce house prices by 35% by the end of the decade. This negative shock and resulting decline in price leaves construction at only 300,000 units by 2000. The eventual steady state of the model requires a construction of 128,000 units to replace losses from the stock. This steady state is reached several years later when prices stabilize at 16.3 (53% of current prices).

In the cycle forecast presented in Table 4, we introduced the 1991 recession into the model. From 1990 to 1991, employment dropped by 1 million jobs, income remained flat, and single family construction dropped to 840,000 units. Our model reasonably tracks this decline, predicting 1991 construction at 879,000 units, only 4.6% higher than actual construction. The cycle forecast of construction for 1991 is almost 28% lower than the base forecast. The model predicts a 29.4% rise in construction for 1992 with construction continuing to rise through 1994. Construction then declines steadily for the rest of the period. By 2000, the stock of single family units in the cycle forecast is 0.7% lower than the base forecast. In the cycle forecast, house prices are initially flat and then rise with the recovery. By 2000, house prices rise 6.2% in real terms over 1990 levels, which is quite similar to the rise over the decade in the base forecast.

VIII. CONCLUSIONS

This paper refines our understanding of the aggregate behavior of the housing market. We present very strong evidence of a gradual price adjustment process in the market for single family housing in contrast to previous assumptions of instantaneous market clearing. Our demand specification provides aggregate estimates of income and price elasticities for

unit demand (or tenure choice). These elasticities are noticeably lower than previous estimates of the demand elasticities for housing services. Our model also suggests that homeownership demand is more sensitive to housing price levels and less sensitive to annual user costs measures than theoretical models would suggest.

On the supply side of the market, we have argued that Muth's stock adjustment model is in fact more in tune with recent housing microeconomics than many other studies. With this model, our long run price elasticity of housing supply is at least 1.2. Annual movements in house prices, however, are too small to explain the large short run swings in single family construction. Much of the yearly variation in new building is better explained by fluctuation in interest rates and other variables besides price that reflect the state of the housing market. The importance of these variables indicates that with slow adjustment, housing prices are not a sufficient statistic for the market. All these results strongly reinforce the notion that the housing market operates in a manner far different from that for other financial assets.

Finally, the forecasts that we present indicate that the much discussed long term crash in the housing market is quite unlikely. To be sure, there may be short term downturns in local markets over the next decade. However, a systematic national depression in single family house prices is extremely improbable. Our model suggests that while demographic forces will slow house price appreciation, the impact will be relatively small. A long run price elastic supply of housing dampens much of the impact of changing demographics. Only in the dour and very unlikely case of no household or income growth over the next decade does our model produce the kind of real declines in house prices predicted by Mankiw and Weil [26].

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